# From "Can We Perceive High-Level Properties?" to "What Makes a State Perceptual?"

A Constraint-Based Reframing of the Range of Perception Debate

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February 2026

**Abstract**

Debates about the perceivability of higher-level properties have remained resiliently inconclusive since the resurgence prompted by Siegel [7]. I argue that the impasse is not primarily empirical but methodological: the debate presupposes an unstable low-level/high-level distinction and lacks a principled criterion for individuating perception *as such*. The central question should be reframed. Rather than asking whether a candidate higher-level property is perceived, we should ask what constraints a mental state must satisfy to count as perceptual rather than as fast judgment, implicit inference, or post-perceptual appraisal. I propose four jointly testable constraints—environmental anchoring, stability under intervention, online action-coupling, and a distinctive error profile—and defend a concise *joint sufficiency* argument: satisfying all four collapses the explanatory space for treating the state as merely "fast inference." I apply the framework to several candidate higher-level cases and to two controls: a positive control (color) and a negative control (semantic provenance). Finally, I show why cognitive penetrability and affective colouring are largely orthogonal to "level," and how cognitive science can constrain mechanisms without turning anatomical localization into a definition.

## 1 Introduction: why the debate keeps failing to conclude

Fifteen years after Siegel [7] sharpened the agenda with phenomenal contrast and a defense of rich perceptual content, debates about higher-level perceivable properties remain surprisingly resistant to closure. The stall has a familiar shape. One camp insists that we *seem* to see causation, animacy, emotion, intention, category membership, danger, and aesthetic value; another insists that such "seeing" is a cognitive overlay—fast judgment, implicit inference, or affective appraisal.

The recurrent dialectical loop is by now predictable: phenomenal contrast motivates a rich-content reading; the skeptic replies with attention, expectation, categorization, or appraisal; the defender replies that these influences can still be perceptual. The debate cycles because it lacks a principled way to distinguish *state-type* (perceptual vs cognitive) from *content-type* (low-level vs higher-level).

This paper offers a diagnosis and a remedy. The diagnosis is that the debate has been asking the wrong *type* of question: it treats *content-type* as if it settled *state-type*. But content-type is not stable enough to be foundational, and phenomenal contrast is not strong enough to carry individuation

by itself. The remedy is a reframing: assess not whether property $P$ is perceptible, but whether the state ascribed to a subject in grasping $P$ satisfies constraints characteristic of perception as a *world-guided, action-relevant, systematically fallible* kind of state.

A crucial clarification: the constraints proposed below are offered as a methodological individuation scheme for the present debate, not as an a priori analysis of the essence of perception. They aim to make commitments testable and disagreements locatable.

## 2 Positioning: what this reframing is (and is not)

Because this paper shifts the central question, it is worth stating how the move relates to three nearby literatures.

First, debates about *consciousness, access, and the format of perceptual representation* ask how much is present in perceptual consciousness relative to what is cognitively accessible for report and reasoning [1]. The present proposal does not attempt to settle that dispute. Instead, it isolates an orthogonal issue: what constraints must hold for any candidate state to qualify as perceptual (as opposed to being an efficient cognitive classifier that happens to feel immediate). One can remain neutral on overflow while insisting on constraint-based individuation.

Second, capacity-based approaches stress that perceptual experience is structured by perceptual capacities and that experience plays an evidential role [6]. The present framework is sympathetic to that orientation but differs in emphasis: instead of taking capacities as primitive explanatory posits, it treats them as hypotheses that should be warranted by operational constraints on state–world coupling.

Third, predictive-processing and action-oriented approaches stress the integration of perception and action, often casting perception as inference under a generative model and emphasizing control-theoretic profiles [2]. The present proposal does not presuppose any single architecture. It agrees that action-relevance matters, but argues that action-coupling alone is too permissive unless paired with intervention tests and distinctive error signatures.

## 3 Why "low-level" vs "high-level" is a shaky starting point

The low-level/high-level distinction is routinely treated as if it were well-behaved. It is not. In the literature, "low-level" has been used to mean:

(a) **stimulus-proximal features** (color, luminance, local edge orientation),

(b) **early-vision outputs** (as in modularity/encapsulation frameworks; [3, 5]),

(c) **non-conceptual contents** (contents not requiring concept possession),

(d) **phenomenal primitives** (what is allegedly phenomenally basic).

These notions come apart. A property can be stimulus-proximal yet concept-involving, or concept-free yet implemented in recurrent, late, distributed processing; it can be phenomenally vivid yet

not bounded by early-vision modules. Consequently, a question like "Is intention high-level?" is ill-posed unless we stipulate *which* sense of "level" is in play.

Even worse, the debate often relies on a tacit slide:

$$\text{high-level content} \Rightarrow \text{cognitive involvement} \Rightarrow \text{not perceptual.}$$

But cognitive involvement and perceptual status are not the same dimension, as the cognitive penetrability literature makes vivid [3, 5, 8, 9]. If "level" is not stable, it cannot be the master key to the range of perception.

## 4 Phenomenal contrast is evidential, not decisive

Siegel-style phenomenal contrast arguments have a clear structure: choose two subject-states that differ phenomenally in a controlled way, infer a difference in content, and argue that the content difference is perceptual [7]. The method is powerful because it forces theory to respect lived phenomenology.

The problem is not that phenomenal contrast provides no evidence. The problem is that it *underdetermines* the state-type that explains the contrast. There are multiple live explanations:

(i) **attentional reconfiguration** (selection, enhancement, suppression),

(ii) **expectation or prior adjustment** (top-down biasing of ambiguous input),

(iii) **learned categorization** (expertise-driven classification),

(iv) **affective salience** (value-tagging that alters what stands out),

(v) **post-perceptual judgment** that feels immediate.

Many of these mechanisms can alter phenomenology without implying that the new content is perceptual in the relevant sense. The methodological moral is sharp: phenomenal contrast is a *screening tool* for candidate contents, not a stand-alone individuation criterion.

## 5 Reframing: from property-type to state-type

The guiding thought is simple:

> *The question "Is property P perceived?" is downstream of the question "What makes a mental state perceptual?"*

Debates about the range of perception implicitly presuppose an answer to the second question; they then fight over the first. That order should be reversed.

Formally, let $\mathsf{W}$ be a space of environmental conditions (scene structure, agent position, occluders, lighting, social cues), $\mathsf{A}$ a space of available actions, and $\mathsf{S}$ the space of candidate internal states attributed to subjects in a task. Let $P$ be a candidate property. The debate typically proceeds:

$$\text{If subjects are in } \mathsf{S} \text{ when exposed to } \mathsf{W}, \text{ does } \mathsf{S} \text{ represent } P?$$

My proposal is to ask instead:

Does $\mathsf{S}$ satisfy constraints that individuate perceptual states for the purposes of this debate?

If yes, then (and only then) we can responsibly ask whether the content of $\mathsf{S}$ includes $P$.

# 6 A constraint-based criterion for perceptual status

This section introduces four constraints. Each is motivated independently, but the aim is joint sufficiency: no single constraint is enough to individuate perception against fast inference in all cases. The constraints are meant to be *operationally constraining*: each has a corresponding family of empirical signatures, while remaining neutral about any single grand theory of vision.

## 6.1 Constraint 1: environmental anchoring

**Definition 1** (Anchoring)**.** *A state* $\mathsf{S}$ *is* environmentally anchored *to property-candidate* $P$ *iff changes in* $\mathsf{W}$ *that are constitutively relevant to* $P$ *induce systematic, non-arbitrary changes in* $\mathsf{S}$*, holding fixed irrelevant background factors.*

**Testable prediction.** Manipulate the relevant environmental variables for $P$ while matching low-level confounds; $\mathsf{S}$ should track the manipulation with predictable shifts, not merely with post-hoc rationalizations.

## 6.2 Constraint 2: stability under intervention

**Definition 2** (Intervention stability)**.** *A state* $\mathsf{S}$ *is* stable under intervention *iff under controlled interventions* $\mathsf{I}$ *that change the causal basis for* $P$ *(while holding constant correlational proxies),* $\mathsf{S}$ *changes in the direction licensed by the intervention rather than by superficial correlations.*

**Testable prediction.** Under interventions that dissociate a proxy cue from the target property (e.g., preserving motion energy while breaking causal contact), a perceptual state should follow the property-relevant intervention, not the proxy.

## 6.3 Constraint 3: online action-coupling

**Definition 3** (Action-coupling)**.** *A state* $\mathsf{S}$ *is* online action-coupled *iff it enters into real-time control of action policies in ways that are sensitive to short-horizon feedback, rather than merely supplying a reportable judgment.*

**Testable prediction.** Under time pressure or dual-task conditions that suppress explicit judgment, performance should covary with $\mathsf{S}$ if $P$ matters for action selection.

### 6.4 Constraint 4: a distinctive error profile

**Definition 4** (Error profile)**.** *A state* $\mathsf{S}$ *has a* distinctive perceptual error profile *iff it exhibits systematic errors under perceptual stressors (e.g., masking, adaptation, crowding, temporal binding windows) that differ from the characteristic errors of belief revision or explicit reasoning.*

**Testable prediction.** Under perceptual stressors, the candidate property should show graded distortions and systematic misbindings, not only categorical flips typical of deliberative appraisal.

### 6.5 Joint sufficiency and non-redundancy

The constraints are not offered as an a priori definition, but they are meant to jointly *exclude* the standard alternative explanation in this debate: that the relevant state is merely fast inference, categorization, or post-perceptual appraisal.

**Proposition 1** (Constraint-based individuation as explanatory exclusion)**.** *For the purposes of the present debate, if a candidate state* $\mathsf{S}$ *(in a given task-domain) is (i) anchored to* $P$*, (ii) stable under interventions that dissociate* $P$ *from typical proxies, (iii) recruited for online action control under time pressure, and (iv) exhibits perceptual-style error signatures under canonical perceptual stressors, then treating* $\mathsf{S}$ *as "merely fast inference" ceases to be an explanatory alternative: any such account must effectively posit a perception-like mechanism (property tracking, online control integration, and perceptual bottlenecks), in which case* $\mathsf{S}$ *counts as perceptual under this individuation scheme.*

**Sketch of the exclusion argument.** Anchoring plus intervention stability jointly force *property tracking*: $\mathsf{S}$ does not merely latch onto a cheap correlational cue, but follows the property-relevant structure across proxy-breaking manipulations. Action-coupling then forces *online control integration*: $\mathsf{S}$ is not a post-hoc add-on to a completed percept; it is recruited in real-time selection and correction of behavior under feedback. Finally, a perceptual error profile forces *perceptual bottlenecks*: graded distortions and misbindings under masking, crowding, adaptation, or timing windows indicate that $\mathsf{S}$ is constrained by the same kinds of resource limits and transformations characteristic of perceptual processing. A pure semantic-knowledge or deliberative-reasoning mechanism would not naturally inherit these signatures except by being functionally and mechanistically yoked to perceptual processing. Thus, once all four constraints are satisfied, the "fast inference" story either (a) fails at least one constraint, or (b) redescribes itself into a perception-like mechanism, collapsing the purported contrast.

**Diagrammatic summary.** Figure 1 provides a compact schematic of how the constraints jointly narrow the explanatory space.

**Four constraints on candidate state $\mathsf{S}$ for property $P$**
(1) Anchoring: $\mathsf{W} \to \mathsf{S}$ tracks $P$-relevant structure
(2) Intervention stability: $\mathsf{I}$ breaks proxies, $\mathsf{S}$ still tracks $P$
(3) Action-coupling: $\mathsf{S}$ enters online control under time pressure
(4) Error profile: $\mathsf{S}$ shows perceptual-style distortions under stressors

$\Downarrow$

**Explanatory exclusion (Proposition 1)**
"Merely fast inference" must either violate at least one constraint, or re-describe itself into a perception-like mechanism (property tracking + online control + perceptual bottlenecks), collapsing the contrast.

Figure 1: Schematic of the constraint package and how it excludes the "merely fast inference" alternative as an explanatory competitor.

### 6.6 A compact evidence map

To reduce "framework talk" to commitments, Table 1 lists common kinds of evidence and which constraints they primarily bear on.

| **Evidence / design family** | **Primary constraint(s) tested** |
|---|---|
| Matched stimulus manipulations (hold low-level confounds fixed) | Anchoring |
| Causal dissociation manipulations (proxy cue vs target property) | Intervention stability |
| Time pressure / dual-task / response deadline paradigms | Action-coupling (with Anchoring) |
| Perceptual stressors (masking, crowding, adaptation, timing windows) | Error profile |
| Cross-context generalization (environmental shift, perturbation robustness) | Anchoring + Intervention stability |
| Mechanistic constraints (distributed vs localized; recurrent vs feedforward) | Mechanism constraints (not definitional), supports all indirectly |

Table 1: A minimal map from evidence types to the constraints they most directly bear on.

## 7 Consequences for familiar confusions

### 7.1 Cognitive penetrability is largely orthogonal to "level"

The cognitive penetrability debate asks whether cognition can directly and nontrivially influence perception [3, 5, 8, 9]. A popular mistake is to treat cognitive influence as evidence that the state is not perceptual. But influence and status can come apart.

A state can satisfy all four constraints (Section 6) while still being modulated by expectation, memory, or concept possession. Conversely, a rapid judgment can be cognitively "sticky" once triggered, yet still not be perceptual. Attention-related complications further muddy simplistic penetration diagnostics [4, 11].

### 7.2 Affective colouring does not settle perceptual status

Affective colouring is often taken to blur perception and cognition: if you see a face as *menacing*, is that perception or appraisal? On the present approach, affective influence is neither a defeater nor a guarantee. It matters only insofar as the resulting state satisfies anchoring, intervention stability, action-coupling, and error-signature constraints. For discussion of evaluative and aesthetic properties as candidates for perceptual representation, see Stokes [10].

## 8 Case studies: disciplined "case-by-case" assessment

This section sketches how the framework evaluates candidate properties without collapsing into ad hoc verdicts. The aim is a reusable template, not a quick verdict.

### 8.1 Causation

Many report *seeing* causation in launch events, contact dynamics, or temporal-contiguity scenes. A classic skeptic says: you see motion and infer causation. The framework asks:

(a) **Anchoring:** does $\mathsf{S}$ systematically track causal manipulations, not merely correlation?

(b) **Intervention stability:** does it follow proxy-breaking interventions (break contact, preserve motion energy)?

(c) **Action-coupling:** does it guide anticipatory action online?

(d) **Error profile:** does it show timing-window or adaptation-like signatures?

### 8.2 Animacy and agency

Seeing animacy in point-light displays or attributing agency to motion patterns is a central candidate. Here action-coupling is especially diagnostic: agency perception should support rapid interactional policies beyond what delayed explicit judgment would permit. Intervention stability is decisive: preserve self-propelled motion cues while disrupting agent-like contingency, and test whether $\mathsf{S}$ follows the property-relevant manipulation.

### 8.3 Intention

Intention attribution is a paradigmatic "too high-level" candidate. The framework does not pre-judge. It asks whether the attributed state is anchored to intention-relevant structure (goal-directedness under perturbation), stable under interventions (decoy goals, obstacle re-planning), and exhibits distinctive error signatures. If the state behaves more like narrative inference (high context variance, low intervention stability), it fails perceptual status.

### 8.4 Affordances

Affordances are action-involving by nature; thus action-coupling is easy to satisfy. The harder tests are anchoring and intervention stability: does the affordance state track objective action-possibility under controlled perturbations, or does it track proxy cues (conventions, size illusions)? Affordances illustrate why no single constraint is sufficient.

### 8.5 Affective properties (danger, disgust, trustworthiness)

These cases invite affect–cognition confusion. The framework forces hard questions: do these states show perceptual-style errors (graded distortions under perceptual load) or do they behave like appraisals (categorical flips driven by background beliefs)? Affective cases are also ideal for showing why penetrability and "level" are orthogonal.

### 8.6 Pictorial perception

Pictorial perception stresses naive individuation: one can grasp objects not present and properties not instantiated in the scene. Here anchoring can be satisfied by stimulus structure: pictorial cues robustly govern experience. Action-coupling may be weaker in ordinary contexts but can be tested via tasks requiring online interaction with pictorially specified layouts. The key is that "not instantiated" does not entail "not perceptual": perception is individuated by constraints on the state, not by metaphysical presence.

### 8.7 Positive control: color as a paradigmatic perceptual property

A framework that only handles controversial cases is under-validated. Consider color as a paradigmatic low-level perceptual property. On the present framework the classification is straightforward: color experience is **anchored** to stimulus conditions (spectral and context-dependent structure), **stable under intervention** in the relevant sense (systematic response to controlled changes in illumination and surface reflectance), **action-coupled** (online sorting, reaching, and discrimination under time pressure), and exhibits a distinctive **perceptual error profile** (e.g., adaptation and context-driven distortions). Thus the framework yields the expected "perceptual" verdict on a canonical case.

### 8.8 Negative control: semantic provenance properties

A framework that only delivers "maybe perceptual" verdicts is too permissive. To test discriminative validity, consider a semantic provenance property: *being Kant's desk*. A subject may visually encounter an antique desk and judge (perhaps correctly) that it is Kant's. But that property is not plausibly perceptual.

On the present framework, the verdict is clear. **Anchoring** fails: changing historical provenance while holding fixed visible features would not systematically alter visual state. **Intervention stability** fails: interventions on provenance (ownership history) do not produce perceptual signatures; at

most, they alter beliefs. **Action-coupling** fails in the relevant sense: whatever action relevance the judgment has is mediated by semantic knowledge and inference, not by an online perceptual control state. **Error profile** fails: mistakes here are the mistakes of memory, testimony, and historical inference, not of masking, crowding, adaptation, or temporal binding. Thus the framework yields a clear non-perceptual classification, as it should.

## 9 What cognitive science can and cannot settle

A tempting empirical shortcut is: if a candidate property is not implemented in a brain area devoted to perception, it is *eo ipso* non-perceivable. This shortcut should be resisted.

First, localization is not individuation. A state can be perceptual by satisfying the constraints above even if its implementation is distributed; conversely, a state implemented in a traditionally perceptual area can still be post-perceptual if its functional role is appraisal-like. Second, perception is not cleanly separable from attention, memory, and expectation in real organisms; what matters is *how* a state is coupled to world-structure and action, and how it fails. Third, cognitive science remains indispensable: it constrains mechanisms and supports intervention tests and error-profile diagnostics [2, 4, 11].

## 10 Contributions and payoffs

This paper contributes a unified methodology for the range-of-perception agenda:

1. It shows why the low-level/high-level distinction is too unstable to ground the debate.
2. It rehabilitates phenomenal contrast as evidence while denying it the status of a criterion.
3. It supplies four jointly testable constraints for perceptual status, applicable across cases.
4. It defends a compact joint-sufficiency argument as *explanatory exclusion* (Proposition 1).
5. It provides both a positive control (color) and a negative control (semantic provenance) to demonstrate discriminative validity.
6. It clarifies that cognitive penetrability and affective colouring are largely orthogonal to "level."
7. It explains how cognitive science can constrain the debate without collapsing it into localization claims.

## 11 Conclusion

The question "Can we perceive higher-level properties?" has persistently failed to yield convergence because it treats content-type as if it settled state-type. The central task is instead to individuate perception. I have proposed a constraint-based framework that forces candidate cases to earn

perceptual status by satisfying environmental anchoring, stability under intervention, online action-coupling, and a distinctive error profile. The package does not trivialize the debate; it makes it harder in the right way. It shifts attention from verbal disputes to testable commitments about how states relate to the world and guide action. If the range-of-perception debate is to mature, it must stop asking merely what we *seem* to see and start asking what makes such seeming genuinely perceptual.

**Acknowledgments.** None.

## References

[1] Ned Block. Perceptual consciousness overflows cognitive access. *Trends in Cognitive Sciences*, 15(12):567–575, 2011.

[2] Andy Clark. *Surfing Uncertainty: Prediction, Action, and the Embodied Mind*. Oxford University Press, Oxford, 2016.

[3] Jerry A. Fodor. *The Modularity of Mind*. MIT Press, Cambridge, MA, 1983.

[4] Steven Gross. Cognitive penetration and attention. *Frontiers in Psychology*, 8:221, 2017. doi: 10.3389/fpsyg.2017.00221.

[5] Zenon W. Pylyshyn. Is vision continuous with cognition? the case for cognitive impenetrability of visual perception. *Behavioral and Brain Sciences*, 22(3):341–365, 1999. doi: 10.1017/S0140525X99002022.

[6] Susanna Schellenberg. *The Unity of Perception: Content, Consciousness, Evidence*. Oxford University Press, Oxford, 2018.

[7] Susanna Siegel. *The Contents of Visual Experience*. Oxford University Press, Oxford, 2010.

[8] Dustin Stokes. Perceiving and desiring: A new look at the cognitive penetrability of experience. *Philosophical Studies*, 158(3):479–92, 2012. doi: 10.1007/s11098-010-9688-8.

[9] Dustin Stokes. Towards a consequentialist understanding of cognitive penetration. In John Zeimbekis and Athanassios Raftopoulos, editors, *The Cognitive Penetrability of Perception: New Philosophical Perspectives*, pages 75–100. Oxford University Press, 2015.

[10] Dustin Stokes. Rich perceptual content and aesthetic properties. In Anna Bergqvist and Robert Cowan, editors, *Evaluative Perception*, pages 19–41. Oxford University Press, 2018.

[11] Petra Vetter, Stephanie Badde, Elisa Ferrè, Janina Seubert, and Barbara Shinn-Cunningham. Evaluating cognitive penetrability of perception across the senses. *Nature Reviews Psychology*, 3, 11 2024. doi: 10.1038/s44159-024-00382-1.